

PolyAgora Multi-Sleeve Alpha Architecture

Layer 0 — Structural Carry Backbone

Stable low-turnover return engine.

Sleeve A — Strategic Carry

Current preferred candidate:

- VIX Roll / Contango Carry
- Futures carry
- FX carry

You already identified this as the stable anchor block.

Target:

- 5–8% exposure baseline
- funded from cash
- always-on unless vetoed

Purpose:

- baseline convex compounding
 - stabilizes Sharpe
-

Layer 1 — Primary Alpha Sleeves

These generate actual edge.

Sleeve B — Cross-Asset Momentum

Most important institutional sleeve.

Types

- 12–1 momentum
- time-series momentum
- cross-sectional momentum
- dual momentum

Assets:

- equities
- rates
- commodities

- FX
- crypto optional

Why important:

Momentum is one of the few persistent market anomalies.

But:

Momentum crashes during transitions.

This is exactly where PolyAgora has advantage:

MRTP + ADD can suppress crash regimes.

This sleeve probably contributes most raw Sharpe.

Sleeve C — Mean Reversion

Essential complement to momentum.

Signals

- z-score reversions
- dispersion reversion
- volatility overextension
- intraday exhaustion
- spread compression

Works best:

- sideways/choppy regimes
- post-panic stabilization

Important:

Momentum + mean reversion are naturally anti-correlated.

This is extremely valuable.

Sleeve D — Relative Value / Spread Arbitrage

Institutional-quality sleeve.

Examples:

- yield curve spreads
- gold/silver
- Brent/WTI

- sector spreads
- ETF/futures basis
- volatility term structure
- bond-equity dislocations

Characteristics:

- market neutral
- lower beta
- high Sharpe
- lower CAGR individually
- excellent portfolio stabilizer

This sleeve is critical for Dov-level benchmarks.

Sleeve E — Volatility / Convexity Sleeve

Very important.

This is what protects institutional systems from regime rupture.

Components

- long convexity overlays
- VIX call structures
- skew positioning
- crisis convexity
- gamma-sensitive overlays

Purpose:

Not constant return.

Purpose:

- prevent catastrophic DD
- preserve convexity during shocks
- allow aggressive deployment after crashes

This is likely missing in V7.8.

Layer 2 — Adaptive Tactical Sleeves

Smaller but powerful.

Sleeve F — Regime Rotation

Rotation across:

- value
- growth
- defensives
- commodities
- inflation trades
- duration

PolyAgora naturally suited for this.

This is where:

- Local Stars
 - admissibility
 - block coherence
- become useful.
-

Sleeve G — Liquidity / Flow Sleeve

Advanced but powerful.

Signals:

- dealer gamma
- CTA positioning
- ETF flow imbalance
- funding stress
- options positioning

This sleeve often predicts short-term transitions before price.

Very compatible with:

- buffer logic
 - ridge detection
 - MRTP steering
-

Sleeve H — Macro Event Sleeve

Optional discretionary/systematic hybrid.

Examples:

- FOMC
- CPI
- payrolls
- geopolitical shocks
- election cycles

Low capital allocation.

High convexity potential.

Layer 3 — Governance Layer (PolyAgora)

This is your moat.

Other funds:

optimize sleeves.

PolyAgora:

governs admissibility of sleeves.

This is the key distinction.

What PolyAgora Should Control

1. Admissibility

Should sleeve exist now?

Not:

“How large is alpha?”

But:

“Is this sleeve coherent with current geometry?”

2. Correlation Governance

Prevent sleeve crowding.

Target:

dynamic low-correlation ensemble.

This is probably the single biggest path toward Dov benchmarks.

3. Transition Suppression

Most quant systems fail here.

Momentum crashes.

Carry explodes.

Mean reversion dies.

PolyAgora can:

- detect ridge zones
- reduce leverage
- steer transitions
- veto incoherent sleeves

This is your strongest theoretical advantage.

4. Kelly Intensity

As you correctly identified:

Governance first.

Kelly second.

Kelly should size:

- only stable admissible edges
 - never define admissibility itself
-

Suggested Institutional Allocation

Stable Version

- 25% Carry
- 25% Momentum
- 15% Mean Reversion
- 15% Relative Value
- 10% Convexity
- 10% Tactical

Could plausibly reach:

- Sharpe 1.1–1.4

Advanced PolyAgora Version

Dynamic allocation:

- MRTP steering
- ridge suppression
- Local-Star activation
- adaptive convex overlays
- sleeve veto

This is where:

Sharpe 1.5+ becomes plausible.

Not guaranteed.

But plausible.

Most Important CTO Insight

Your architecture should evolve toward:

“Governed ensemble intelligence.”

Not:

- one master strategy
- one optimizer
- one signal

Instead:

many imperfect sleeves,
governed coherently under changing geometry.

That is institutionally credible.

ok - full spect to my CTO next level

PolyAgora V8 — Multi-Sleeve Alpha Architecture

CTO Specification — “Governed Ensemble Intelligence”

1. Core Objective

Transform PolyAgora from:

“single governed strategy”

into:

“governed multi-sleeve alpha architecture”

The objective is NOT merely:

- lower drawdown,
- smoother equity,
- or better gating.

The objective is:

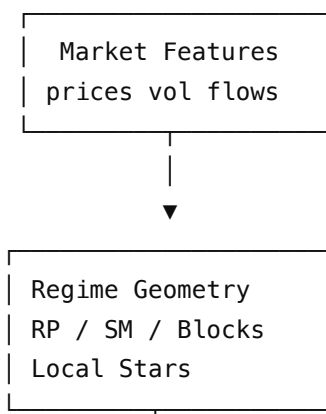
Create a dynamically governed ensemble of partially independent alpha engines operating under Fixed-Star / MRTP admissibility logic.

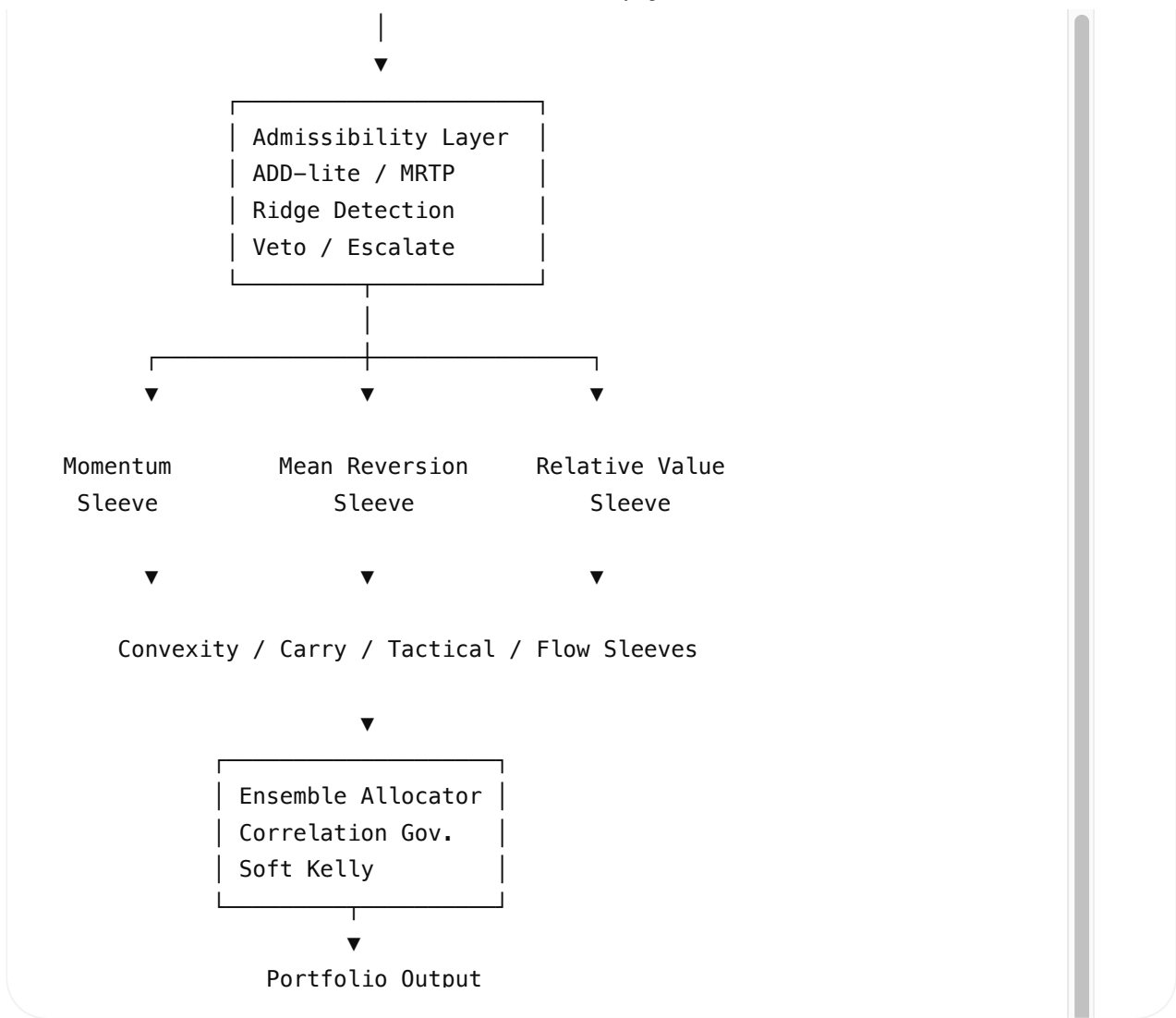
This is the required direction to approach:

- Sharpe > 1.2–1.5
- Sortino > 1.5–2.0
- Calmar > 1.0–1.5
- DD < 8–10%

Institutional target.

2. High-Level Architecture





3. Philosophy

Existing V7.x

Current architecture:

- strong governance,
- weak alpha diversity.

Result:

- excellent DD suppression,
- moderate CAGR.

Problem:

- insufficient independent return streams.

V8 Principle

PolyAgora should:

- govern sleeves,
- not replace sleeves.

The alpha comes from:

- sleeves.

The robustness comes from:

- PolyAgora governance.

4. Sleeve Layer

Sleeve A — Momentum Engine

Goal

Primary trend capture.

Variants

Implement modularly:

- Time-Series Momentum
- Cross-Sectional Momentum
- Dual Momentum
- Relative Momentum

Signals

Time-Series

```
signal_i = sign(ret_12m_i)
```

Cross-Sectional

```
rank_i = percentile_rank(ret_12m_i)
```

Dual

```
if asset_return > cash_return:
```

Features

- medium-term trend
- persistent anomalies
- crisis participation

Weakness

Momentum crashes:

- sharp reversals,
- transition regimes.

PolyAgora role

Suppress during:

- ridge instability,
- star incoherence,
- regime collapse.

Sleeve B — Mean Reversion Engine

Goal

Exploit overshoot/reversion.

Signals

- z-score
- realized vol expansion
- Bollinger stretch
- RSI exhaustion
- spread dislocation

Example

```
z = (price - rolling_mean)/rolling_std  
signal = -z
```

Role

- anti-correlated to momentum
 - stabilizes ensemble
-

Sleeve C — Relative Value Engine

Goal

Market-neutral structural alpha.

Examples

- yield curve spreads
- sector spreads
- ETF basis
- commodities spreads
- volatility term structure

Important

This sleeve:

- often low CAGR individually,
- but extremely high Sharpe contribution.

Critical for:

- Dov benchmarks.
-

Sleeve D — Carry Engine

Goal

Structural carry extraction.

Instruments

- VIX roll yield
- futures carry
- FX carry
- curve carry

Important

Always-on baseline unless:

- veto,

- convexity stress,
 - ADD rupture.
-

Sleeve E — Convexity Engine

Goal

Protect against:

- regime rupture,
- tail events,
- volatility explosions.

Components

- VIX calls
- skew overlays
- gamma structures
- crisis convexity

Allocation

Normally:

- small negative carry.

During stress:

- portfolio survival amplifier.
-

Sleeve F — Regime Rotation

Goal

Rotate among:

- growth
- value
- duration
- inflation
- defensives
- commodities

Controlled by

- Local Stars

- block admissibility
 - transition scoring
-

Sleeve G — Flow/Liquidity Engine

Advanced

Optional phase-2.

Signals:

- dealer gamma
- CTA positioning
- ETF flows
- funding spreads
- options positioning

Purpose:

detect transition before price.

Very important long-term.

5. Governance Layer (PolyAgora Core)

This is the moat.

Other funds:

optimize alpha.

PolyAgora:

governs admissibility.

6. Admissibility Engine

For each sleeve:

$A_{\text{sleeve}}(t) \in [0, 1]$

Interpretation:

- $0 \rightarrow$ inadmissible
- $1 \rightarrow$ fully admissible

Computed from:

- block coherence,
 - star distance,
 - ridge tension,
 - transition instability,
 - ADD-lite risk.
-

7. Ridge Detection

Critical.

Detect:

- transition instability,
- ordering inconsistency,
- star conflict.

Inputs

- volatility expansion
 - correlation spike
 - regime dispersion
 - SM instability
 - Local-Star disagreement
-

8. Veto Logic

Definition

A sleeve is vetoed if:

- expected recoverability collapses,
- admissibility below threshold,
- transition instability exceeds margin.

Example

```
if admissibility <  $\theta_a$ :
```

9. MRTP Runtime Steering

NOT:

maximize immediate reward.

BUT:

maximize future recoverability.

Runtime rule

Select trajectory maximizing:

`recoverability_score`

rather than:

`local_pnl`

10. Correlation Governance

MOST IMPORTANT NEW LAYER.

Current V7 weakness:

too few independent return streams.

Dynamic Correlation Matrix

Compute:

`Corr_ij(t)`

for all sleeves.

Objective

Avoid:

- sleeve crowding,
 - hidden leverage,
 - correlated collapse.
-

Target

Portfolio-level:

Average sleeve correlation < 0.35

11. Ensemble Allocator

Final weight:

```
w_i =  
admissibility_i  
× alpha_strength_i  
× diversification_score_i  
× Kelly_factor_i
```

normalized globally.

12. Soft Kelly Layer

Kelly is NOT governance.

Kelly only sizes:

- stable admissible edges.
-

Rule

```
kelly_i =  
edge_i / variance_i
```

bounded:

```
0 ≤ kelly_i ≤ k_max
```

Suggested:

```
k_max = 0.25
```

No full Kelly.

13. Convexity Re-Entry

Critical missing component from V7.8.

Current issue:

system suppresses risk but re-enters too slowly.

Add Rule

When:

- ADD risk declines,
- emerging block coherence rises,
- ridge tension falls,

allow:

- progressive convex re-deployment.
-

14. Runtime States

Zone 1 — Stable

- full sleeves active
- Kelly enabled

Zone 2 — Transition

- reduce leverage
- suppress correlated sleeves
- activate MRTP routing

Zone 3 — Stress

- defensive rotation
- convexity increase

Zone 4 — Rupture

- carry veto
 - momentum veto
 - convexity dominant
 - search next Local Star
-

15. Testing Requirements

MANDATORY.

A. Walk-Forward

Never:

single static backtest only.

Must:

walk-forward:

- train
 - validate
 - deploy
-

B. Transition Windows

Explicitly evaluate:

- GFC
 - COVID
 - 2022
 - flash crashes
 - inflation shock
-

C. Correlation Stress

Measure:

Sleeve correlation under stress

D. Regime Transition Accuracy

Track:

- veto precision
- false veto
- late transition
- recoverability preservation

16. Institutional Targets

Initial Target

Sharpe > 1.2
Sortino > 1.5
Calmar > 1.0
DD < 8%
Corr(SPY) < 0.35

Advanced Target

Sharpe > 1.5
Sortino > 2.0
Calmar > 1.5
DD < 8%

17. Most Important CTO Insight

PolyAgora should evolve into:

“a governed ensemble operating under semantic admissibility geometry.”

NOT:

- single optimizer,

- single signal,
- single master strategy.

The future architecture is:

many imperfect sleeves,
dynamically governed,
under recoverability-aware regime geometry.

That is the path toward institutional quality.